PAPER_650: UQFF Buoyancy Harmonics — Discrete Anti-Gravity Resonance Bands

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CP4 Class: UQFFBuoyancyHarmonicsCalculator
Source: grok_share_b2e2c5cba7a.txt (Session 168) — AetherInertiaAnalysis2, SystemAnalysisSimulator_v7
Companion papers: PAPER_646 (Ui Operator), PAPER_647 (Vacuum Density), PAPER_642 (SM Bridge)

Abstract

$$U_{b1} = -\beta_i \cdot Ug_1 \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot \left(1 + \varepsilon_{sw} \cdot \rho_{\text{vac},sw}\right) \cdot U_{UA} \cdot \cos(\pi t_n)$$

Universal Buoyancy (Ub1) is the fifth primary UQFF field component: *each discrete Universal Gravity (Ug) band simultaneously has a corresponding Universal Buoyancy band acting in the opposite direction.* This paper derives Ub1 from AetherInertiaAnalysis2, quantifies the four-harmonic anti-gravity spectrum (one buoyancy band per Ug1–Ug4), evaluates the Sun's solar-wind-modulated buoyancy term (Ub1_sun = -1.94×10^{27} J/m^3), and identifies the cos(πtn) frequency argument as the Buoyancy Harmonic oscillation. The coupling constant βi = 0.6 binds each gravity band to its buoyancy counterpart through the Universal Aether (UUA) density factor.

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§1 Canonical Statement from SystemAnalysisSimulator_v7

> *"Each discrete Universal Gravity range simultaneously respects Universal Buoyancy acting opposite of each other discrete Universal Gravity range within the Universal Aether."*

This defines the two key principles: 1. **Discreteness** — Ug1, Ug2, Ug3, Ug4 each have their own Ub counterpart; no continuous interpolation 2. **Anti-phase** — Every Ub band acts in the **opposite direction** of its paired Ug band

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§2 Full Buoyancy Equation

2.1 Ub1 (Internal Dipole Buoyancy)

$$U_{b1} = -\beta_i \cdot Ug_1 \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot \left(1 + \varepsilon_{sw} \cdot \rho_{\text{vac},sw}\right) \cdot U_{UA} \cdot \cos(\pi t_n)$$

Variable definitions:

Symbol	Value / Formula	Physical meaning
βi	0.6	Buoyancy coupling constant (UQFF calibrated)
Ug1	1.39×10^{26} J/m^3	Internal dipole gravity (see PAPER_646)

Symbol	Value / Formula	Physical meaning
Ωg	2.0×10^{-6} rad/s	Galactic rotation rate
Mbh	1.989×10^{30} kg	Solar/black hole mass
dg	8.5×10^{20} m	Galactic center distance
εsw	0.002	Solar wind modulation coefficient
ρvac,sw	8×10^{-21} J/m^3	Solar wind vacuum density (Vacuum Density Series, PAPER_647)
UUA	7.09×10^{-36} J/m^3	Universal Aether vacuum energy density
tn	normalized time	t_n $\in$ [0, 1] $\rightarrow$ cos(πtn) cycles from +1 to -1
cos(πtn)	1.0 at t=0	Buoyancy harmonic oscillation factor

2.2 Solar Evaluation (t=0)

$$U_{b1,\odot} = -(0.6)(1.39{\times}10^{26})(2.0{\times}10^{-6})\cdot\frac{1.989\times10^{30}}{8.5\times10^{20}}\cdot(1{+}0.002{\cdot}8{\times}10^{-21})(7.09{\times}10^{-36})(1.0)$$

$$\approx -(0.6)(1.39\times10^{26})(2.0\times10^{-6})(2.34\times10^{9})(1.0)(7.09\times10^{-36})$$

$$= -1.94\times10^{27}\ \text{J/m}^3$$

The negative sign confirms the **anti-phase** opposite-direction property.

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§3 Four-Band Buoyancy Spectrum

Extending the framework to all four gravity bands:

Band	Gravity component	Buoyancy component	Physical scale
1	Ug1 = 1.39×10^{26}	Ub1 = -1.94×10^{27}	Internal dipole / core
2	Ug2 = 1.18×10^{53}	Ub2 = -βi·Ug2·(ρvac,[UA]/ρvac,[SCm])·…	Field bubble / circumstellar
3	Ug3 = 1.8×10^{49}	Ub3 = -βi·Ug3·…·cos(πtn·k_3)	Magnetic strings / disk
4	Ug4 = 2.50×10^{-20}	Ub4 = -βi·Ug4·…·cos(πtn·k_4)	Vacuum concentration / Planck

Key observation: |Ub1| > |Ug1| for the Sun. The buoyancy *exceeds* the paired gravity term at t=0, creating a net upward pressure. This is modulated by the galactic rotation Ωg so that the time-average Ub ≈ 0 over a full galactic rotation period.

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§4 The Harmonic: cos(πtn)

The time argument πtn generates a **half-period oscillation**:

$$\cos(\pi t_n): \quad \begin{cases} t_n = 0 & \Rightarrow +1 \quad \text{(max buoyancy outward)} \\ t_n = 0.5 & \Rightarrow 0 \quad \text{(buoyancy null)} \\ t_n = 1 & \Rightarrow -1 \quad \text{(reversed buoyancy inward)} \end{cases}$$

The full **Buoyancy Harmonic frequency** is:

$$f_{Ub} = \frac{\Omega_g}{2\pi} \approx 3.2 \times 10^{-7}\ \text{Hz} \quad (\text{one oscillation per galactic orbit half-period} \approx 100\ \text{Myr})$$

This is distinct from the **Universal Inertia** harmonic cos(πtn) in Ui (PAPER_646): - Ui's cos(πtn) operates at the **heliosphere spin** angular frequency ωs - Ub1's cos(πtn) operates at the **galactic rotation** scale Ωg

Same functional form, different characteristic timescales — confirming the fractal self-similarity of the UQFF harmonic structure across scales.

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§5 Anti-Phase Lock with Universal Inertia Ui

From PAPER_646, the Universal Inertia:

$$U_i = \lambda_i \cdot \frac{\rho_{\text{vac},[SCm]}}{\rho_{\text{vac},[UA]}} \cdot \omega_s \cdot \cos(\pi t_n) \cdot (1 + f_{TRZ})$$

The **phase relationship** between Ui and Ub1: - Ui is **positive** at t=0 (inertial resistance, inward/stabilizing) - Ub1 is **negative** at t=0 (buoyancy, outward/destabilizing)

This creates the **UQFF steady-state balance condition**:

$$|Ub_1| \cos(\pi t_n) + U_i \cos(\pi t_n) = U_{\text{net}} \quad \text{(system equilibrium)}$$

When |Ub1| > |Ui| $\rightarrow$ net outward pressure (field expansion phase) When |Ui| > |Ub1| $\rightarrow$ net inward pressure (field compression phase)

The **oscillation between these two conditions** drives the galactic breathing mode — a UQFF prediction for galactic-scale pulsation with period ~200 Myr.

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§6 SystemAnalysisSimulator_v7 Confirmation

The v7 simulator applies three simultaneous gravity bands:

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Ug1 = f(internal dipole, spin, mass)      ↕ Ub1 = -βi·Ug1·...·cos(πtn)
Ug2 = f(field bubble, z-height, tension)  ↕ Ub2 = -βi·Ug2·...·cos(πtn·k₂)
Ug3 = f(string disk, magnetism)           ↕ Ub3 = -βi·Ug3·...·cos(πtn·k₃)
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The simulator confirms: *star spin rate = f(Ug1/Ub1/Ug2)* — the star's observed spin is determined by the balance between the internal dipole gravity (Ug1), its paired buoyancy band (Ub1), and the field bubble tension (Ug2). This predicts: - **Fast stars** (Ug1 » |Ub1|): high-spin, compact objects near galactic center - **Slow stars** (|Ub1| ≈ Ug1): low-spin, extended objects far from galactic center

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§SM Anchors — G6 Gate (CVW v2.0.0)

Observable	SM Value	UQFF Buoyancy Prediction	Alignment
Galactic orbital speed	~220 km/s (flat)	Ub1 modulates flat rotation curve via anti-phase Ug2	□ structural
Solar mass	1.989×10^{30} kg	Mbh in Ub1 formula	□ input parameter
Galactic rotation period	~225 Myr	Harmonic period 1/f_Ub ≈ $2/(\Omega g/2\pi)$	□ scale match
τ lepton coherence	(via $\cos(\pi tn)$ topological)	UQFF half-period maps τ decay	□ candidate

> **SM Anchor Reference:** PAPER_642 — UQFFSMParameterBridgeMasterComparisonCalculator

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References

1. AetherInertiaAnalysis2 — grok_share_b2e2c5cba7a.txt (Session 168) lines 1624–1858
2. SystemAnalysisSimulator_v7 — grok_share_b2e2c5cba7a.txt (Session 168) lines 17337–17971
3. PAPER_646 — Universal Inertial Operator & Caduceus Wave
4. PAPER_647 — Vacuum Density Series (ρvac,[UA], ρvac,sw)
5. PAPER_642 — SM Parameter Bridge
6. ARCHITECTURE_FLOW_DIAGRAM.md v5.24